

ZILINA HRICOV, Slovakia

WMO: 118410

Lat: 49.233N

Lon: 18.614E

Elev: 311

StdP: 97.64

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.2	-11.5	-17.2	0.9	-13.0	-14.6	1.1	-10.4	7.2	1.4	6.5	1.2	1.5	100	0.506

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	30.8	20.4	28.9	19.6	27.1	18.6	21.2	28.6	20.3	27.0	19.5	25.4	2.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	14.2	24.0	18.0	13.4	22.9	17.2	12.7	21.9	63.1	28.9	59.6	27.4	56.6	25.4	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.7	5.8	5.1	-17.9	33.2	3.5	1.4	-20.4	34.3	-22.5	35.1	-24.5	35.9	-27.0	37.0		
(5)			-18.2	23.0	3.5	1.0	-20.7	23.7	-22.7	24.3	-24.7	24.9	-27.2	25.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	-2.2	-0.3	3.6	9.4	13.9	17.3	19.1	18.8	14.2	9.4	4.6	-0.7		
	DBStd	8.44	4.56	4.30	4.00	4.14	3.55	3.40	3.24	3.19	3.36	3.92	4.24	4.37		
	HDD10.0	1496	377	288	200	58	8	0	0	0	4	60	168	332		
	HDD18.3	3557	636	522	457	268	143	59	30	33	130	277	412	591		
	CDD10.0	1121	0	0	1	40	128	219	283	273	129	42	6	0		
	CDD18.3	141	0	0	0	0	5	28	55	48	5	0	0	0		
	CDH23.3	1596	0	0	0	14	98	335	582	499	65	2	0	0		
	CDH26.7	464	0	0	0	1	12	90	195	158	8	0	0	0		
(14) Wind	WSAvg	1.8	1.6	1.7	2.1	2.1	2.1	2.0	1.9	1.6	1.6	1.5	1.6	1.6		
(15) Precipitation	PrecAvg	791	52	47	50	53	78	87	102	76	71	62	61	53		
	PrecMax	1059	135	103	131	112	207	139	199	184	152	140	119	133		
	PrecMin	557	13	9	11	4	30	30	28	15	21	4	4	17		
	PrecStd	103	29	23	31	25	38	28	43	44	39	38	28	26		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.7	12.0	18.3	25.2	28.1	31.2	33.2	33.2	28.0	22.8	17.9	12.0		
		MCWB	7.1	7.7	10.3	14.4	17.4	20.8	20.6	20.7	18.4	16.2	12.4	8.5		
	2%	DB	7.0	9.1	15.6	22.5	26.0	29.3	30.9	30.3	25.1	20.1	15.0	8.7		
		MCWB	5.5	5.9	8.9	13.2	16.6	19.9	20.4	20.3	17.4	14.7	11.3	6.7		
	5%	DB	5.0	7.0	12.9	20.0	24.0	27.2	28.9	28.2	23.1	17.9	12.9	6.2		
		MCWB	4.0	4.7	7.6	12.0	16.0	18.8	19.5	19.4	16.4	13.6	9.8	5.0		
	10%	DB	3.2	5.1	10.1	17.7	21.9	25.0	26.8	26.1	21.0	15.5	10.8	4.6		
		MCWB	2.3	3.5	6.4	10.8	14.9	17.8	18.7	18.5	15.6	12.4	8.7	3.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.6	8.1	10.7	15.2	18.9	22.0	22.4	22.5	19.6	16.8	13.1	9.0		
		MCDB	9.1	11.0	16.6	23.2	25.3	29.6	30.1	29.2	25.9	21.0	16.9	11.1		
	2%	WB	5.6	6.3	9.5	13.8	17.6	20.6	21.3	21.1	18.3	15.3	11.6	6.5		
		MCDB	6.6	8.4	14.3	20.7	24.1	27.5	28.7	28.5	23.6	19.0	14.6	8.0		
	5%	WB	4.0	4.9	8.1	12.6	16.6	19.6	20.3	20.0	17.1	14.0	10.2	5.1		
		MCDB	4.8	6.5	11.9	18.5	22.5	25.6	26.9	26.9	21.7	17.2	12.6	6.1		
	10%	WB	2.4	3.6	6.7	11.5	15.5	18.5	19.4	19.1	16.1	12.8	8.8	3.7		
		MCDB	3.1	4.9	9.7	16.6	20.6	23.8	24.9	25.0	20.1	15.4	10.4	4.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.7	7.5	9.2	11.6	11.5	11.5	11.7	12.1	10.3	9.1	6.4	4.9		
		MCDBR	6.1	10.7	14.5	16.3	15.7	15.4	15.7	15.5	14.2	12.7	9.9	6.3		
	5% WB	MCWBR	4.8	7.3	8.4	8.1	7.5	7.0	6.4	6.5	7.0	7.5	6.3	4.8		
		MCWBR	5.6	9.2	12.6	14.7	14.0	13.7	13.9	14.3	12.4	11.6	8.9	5.8		
(40) Clear-Sky Solar Irradiance	taub		0.307	0.332	0.376	0.428	0.426	0.428	0.439	0.433	0.403	0.372	0.343	0.306		
		taud	2.303	2.262	2.226	2.161	2.212	2.245	2.218	2.245	2.298	2.361	2.369	2.367		
	Ebn at Noon		734	802	824	817	835	835	818	803	788	750	686	687		
		Edh at Noon	71	94	116	137	136	133	134	125	107	86	68	59		
(44) All-Sky Solar Radiation	RadAvg		0.91	1.66	2.77	4.08	4.87	5.24	5.07	4.59	3.25	1.98	1.01	0.72		
	RadStd		0.15	0.26	0.46	0.48	0.62	0.52	0.57	0.44	0.49	0.28	0.15	0.08		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.52	N/A	N/A	N/A	+0.54	+0.48	-161	-219	+92	+45
(47) Regional (54 neighbors)		+0.53	N/A	N/A	+0.81	+0.43	+0.38	-125	-163	+77	+32

Nomenclature: See separate page